

# MARKOSE TSEGAYE

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## Education

### George Mason University - 3.67 GPA

*B.S Computer Science (Accelerated M.S. Track)*

**Dec 2026**

*Fairfax, VA*

### George Mason University

*Accelerated M.S. in Computer Science*

**Expected Dec 2027**

*Fairfax, VA*

## Experience

### Software Engineer Intern

**May 2026 - Aug 2026**

*Credit Acceptance | Mobile Consumer Experience*

*Remote*

- Built React Native flows for payments and loan details in the new Credit Acceptance app and website for **4M+** users.
- Leveraged agentic AI for migration, code generation, and testing, cutting the rewrite timeline from **18 months to 3**.
- Implemented PII masking, security compliance checks, and secure bank integrations for payment processing flows.
- Integrated RESTful API's to sync account and payment data between the React Native app and Spring Boot backend.

### Software Engineer

**May 2025 - July 2025**

*The Dumpster Masters | Local Waste Management Client*

*Rockville, MD*

- Delivered a React Native MVP for a local trash compaction company, supporting **3** pilot drivers and **100+** jobs weekly.
- Designed a route planning system using mapping APIs and geospatial data, improving operational efficiency by **35%**.
- Implemented a secure Express.js REST API with **20+** endpoints supporting authentication, job scheduling, billing, and admin workflows, handling **200+** daily requests with PostgreSQL-backed transactional integrity.
- Engineered backend support for Dumpster Masters' appointment scheduling, including time-slot locking, cron-driven cleanup jobs, and race-condition protection to prevent double bookings under concurrent booking requests.

### Software Engineer - Startup

**July 2025 - Sept 2025**

*Forkcast | Predictive Restaurant Platform*

*Remote*

- Owned backend data intake and processing pipelines while collaborating with a **5**-member agile team.
- Developed Express and FastAPI ingestion APIs backed by BullMQ workers to asynchronously process **300K+** rows of POS datasets, enabling near real-time inventory trends and recommendations without blocking client requests.

## Projects

### AlgoMinds (*Algominds.net*) | Typescript, Next.js, Express.js, WebSockets, Redis

**March 2025 – Present**

- Built and scaled AlgoMinds, a real-time multiplayer coding platform supporting **100+** concurrent users.
- Led backend development while collaborating with another engineer who handled authentication and frontend integration, ensuring seamless interaction between real-time and user systems.
- Designed and implemented a distributed systems pipeline using Redis queues and asynchronous processing to execute user-submitted code, reducing execution latency by **40%** and preventing system bottlenecks.
- Deployed backend services to Fly.io and Azure, integrating centralized logging with Winston and Sentry error tracking.
- Reduced unhandled runtime failures by **30%** through structured error handling and proper monitoring.

## Leadership

### Event Coordinator - ColorStack GMU

**Oct 2025 – Present**

*George Mason University*

*Fairfax, VA*

- Coordinated logistics for 12+ events per semester including venue booking, university partnerships, catering, and attendance tracking, averaging 20–30 members per event.
- Supported a chapter of 100+ underrepresented CS students at GMU by connecting members with industry partners for internship and mentorship opportunities.

## Technical Skills

**Languages:** TypeScript, JavaScript, Python, Java, C, SQL, HTML/CSS

**Frameworks and Libraries:** React, React Native, Expo, Yarn, Node.js, Next.js, Express.js, FastAPI, Django, BullMQ

**Databases & Cloud:** PostgreSQL, Redis, Supabase, Azure DevOps, AWS, Fly.io, Firebase, Google Cloud Platform

**Platforms & Tooling:** Git, GitHub, VS Code, Android Studio, XCode, Docker, Figma